



Theory Assignment Report :1

Only for course Teacher					
	Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage	25%	50%	75%	100%	20
Clarity					
Content Quality					
Spelling & Grammar					
Organization and Formatting					
Total obtained mark					
Comments					

Semester: Fall 2026

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Batch: 46

Section: B

Course Code: CSE 124

Course Name: Object Oriented Programming Lab Language Lab

Course Teacher Name: Debabarata Mallick

Designation: Lecturer

Submission Date: 30/ 07/ 2026

$$\vdots$$

Hello

Amit Dey

ANSWER:

```
J gc1.java > ...  
1 public class gc1 {  
    Run | Debug  
2     public static void main (String[] args){  
3         System.out.println(x: "Hello");  
4         System.out.println(x: "Amit Dey");  
5     }  
6 }  
7  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: gc1 + v [ ] [ ] ... | [ ] [X]  
  
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat_java_jdt_ws\Java_9d12eabf\bin\' 'gc1'  
Hello  
Amit Dey  
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

Explain:

I learned how to declare a Java class and the `main()` method. I also learned how to use `System.out.println()` to print text on the console.

2. Sum of Two Numbers

Write a Java program to print the sum of two numbers.

Test Data:

 $74 + 36$
$$\vdots$$

110

ANSWER:

```
J gc2.java > gc2 > main(String[])  
1 class gc2{  
    Run | Debug  
2     public static void main(String[] args){  
3         int a = 74;  
4         int b = 36;  
5         int sum = a+b;  
6         System.out.println(sum);  
7     }  
8 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc2'
110
PS C:\Users\amitd\OneDrive\Desktop\Java>

Explain:

I learned how to declare integer variables and store values in them. And I also learned how to add two numbers using the `+` operator and display the result with `System.out.println()`.

3. Division of Two Numbers

Write a Java program to divide two numbers and print them on the screen.

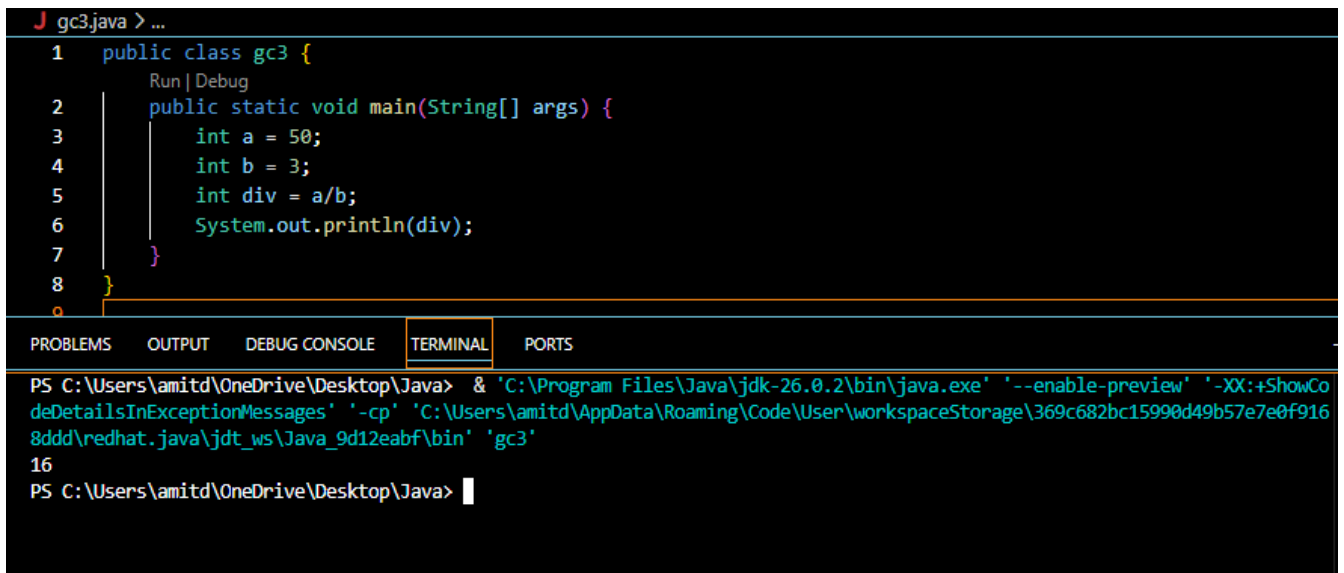
Test Data:

50/3

:

16

ANSWER:



```
J gc3.java > ...
1 public class gc3 {
    Run | Debug
2     public static void main(String[] args) {
3         int a = 50;
4         int b = 3;
5         int div = a/b;
6         System.out.println(div);
7     }
8 }
9

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc3'
16
PS C:\Users\amitd\OneDrive\Desktop\Java> |
```

Explain:

I learned how to perform division using the `/` operator in Java. I learned how to store the result in a variable and print it using `System.out.println()`.

4. Arithmetic Operations

Write a Java program to print the results of the following operations.

a. $-5 + 8 * 6$

b. $(55+9) \% 9$

c. $20 + -3*5 / 8$

d. $5 + 15 / 3 * 2 - 8 \% 3$

:

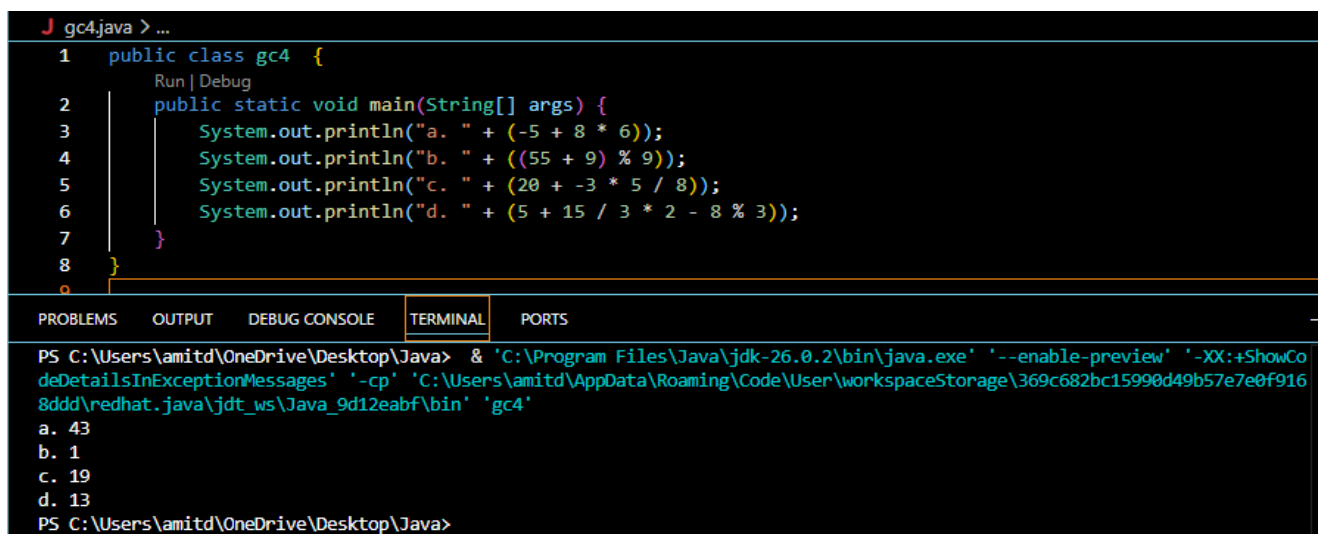
43

1

19

13

ANSWER:



The screenshot shows an IDE with a Java program named 'gc4.java'. The program contains four lines of code that calculate and print the results of the arithmetic operations specified in the problem. The output window shows the results: 'a. 43', 'b. 1', 'c. 19', and 'd. 13'.

```
gc4.java > ...  
1 public class gc4 {  
    Run | Debug  
2     public static void main(String[] args) {  
3         System.out.println("a. " + (-5 + 8 * 6));  
4         System.out.println("b. " + ((55 + 9) % 9));  
5         System.out.println("c. " + (20 + -3 * 5 / 8));  
6         System.out.println("d. " + (5 + 15 / 3 * 2 - 8 % 3));  
7     }  
8 }  
9  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCo  
deDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f916  
8ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc4'  
a. 43  
b. 1  
c. 19  
d. 13  
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

Explain:

I learned how to evaluate arithmetic expressions using operators such as $+$, $-$, $*$, $/$, and $\%$. I learned that Java follows **operator precedence** and evaluates expressions accordingly.

5. Product of Two Numbers

Write a Java program that takes two numbers as input and displays the product of two numbers.

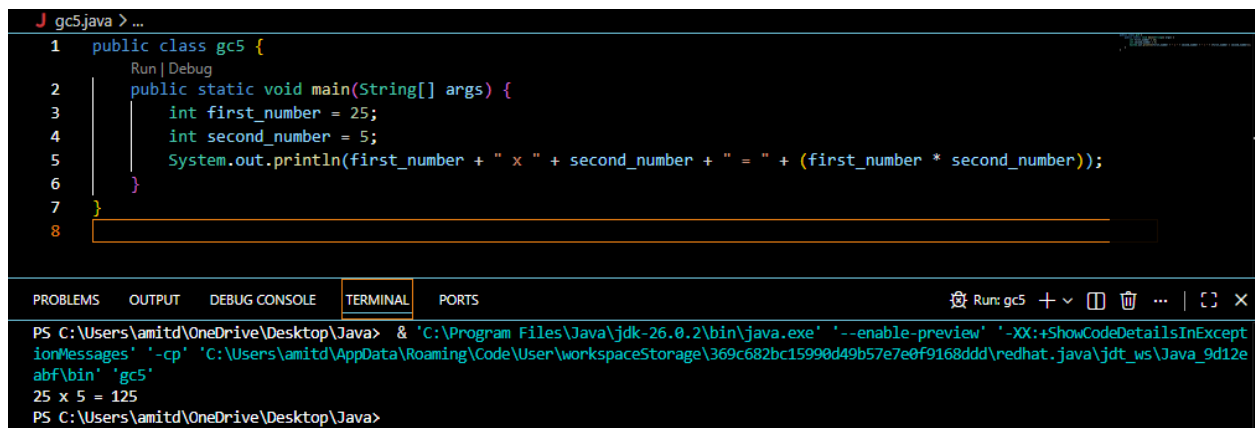
Input first number: 25

Input second number: 5

:

25 x 5 = 125

ANSWER:



The screenshot shows an IDE with a Java file named `gc5.java`. The code defines a class `gc5` with a `main` method. Inside `main`, two integer variables are declared and assigned: `first_number = 25` and `second_number = 5`. A `System.out.println` statement is used to print the product of these two numbers, formatted as `"x" + " = " + (first_number * second_number)`. The IDE's interface includes tabs for `PROBLEMS`, `OUTPUT`, `DEBUG CONSOLE`, `TERMINAL`, and `PORTS`. The `TERMINAL` tab is active, showing the command prompt execution of the program, which results in the output `25 x 5 = 125`.

```
1 public class gc5 {  
2     public static void main(String[] args) {  
3         int first_number = 25;  
4         int second_number = 5;  
5         System.out.println(first_number + " x " + second_number + " = " + (first_number * second_number));  
6     }  
7 }  
8
```

PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc5'
25 x 5 = 125
PS C:\Users\amitd\OneDrive\Desktop\Java>

Explain:

I learned how to declare variables and store numeric values in Java. I learned how to multiply two numbers using the `*` operator and print the result in a formatted output using `System.out.println()`.

6. Basic Arithmetic Operations

Write a Java program to print the sum (addition), multiply, subtract, divide and remainder of two numbers.

Input first number: 125

Input second number: 24

$$\vdots$$
$$125 + 24 = 149$$
$$125 - 24 = 101$$
$$125 \times 24 = 3000$$
$$125 / 24 = 5$$
$$125 \bmod 24 = 5$$

ANSWER:

J gc6.java > ...

Run | Debug

```
1 public class gc6 {  
    Run | Debug  
2     public static void main(String[] args){  
3         int first_number= 125;  
4         int second_number=24;  
5         int sum= first_number + second_number;  
6         int subtract = first_number - second_number;  
7         int multiplication= first_number * second_number;  
8         int division = first_number / second_number;  
9         int modulus = first_number % second_number;  
10        System.out.println("Sum:" +sum);  
11        System.out.println("Subtract:" +subtract);  
12        System.out.println("multiplication:" +multiplication);  
13        System.out.println("division:" +division);  
14        System.out.println("modulus:" +modulus);  
15    }  
16 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Run: gc6

+ v

...

X

```
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e70f9168ddd\redhat.java_jdt_ws\Java_9d12eabf\bin\' 'gc6'  
Sum:149  
Subtract:101  
multiplication:3000  
division:5  
modulus:5  
  
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

Explain:

I learned how to store the results of arithmetic operations (+, -, *, /, %) in separate variables. I learned how to display multiple calculated results using `System.out.println()`.

10. Formula Computation

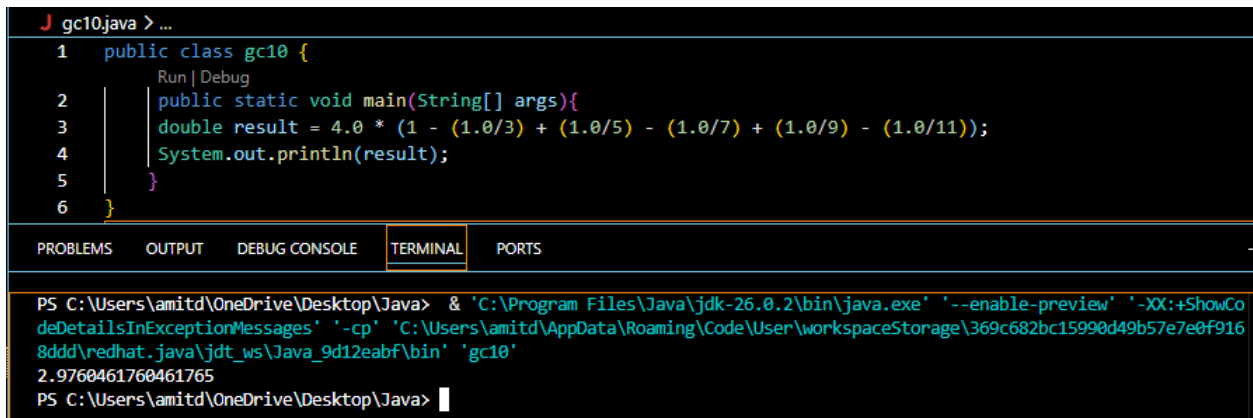
Write a Java program to compute a specified formula.

$4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11))$

:

2.9760461760461765

ANSWER:



The screenshot shows an IDE with a Java file named `gc10.java`. The code defines a public class `gc10` with a `main` method. Inside the `main` method, a `double` variable `result` is calculated using the formula $4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11))$, and the result is printed to the console. The IDE interface includes tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the command to run the program and the output `2.9760461760461765`.

```
J gc10.java > ...
1 public class gc10 {
    Run | Debug
2     public static void main(String[] args){
3         double result = 4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11));
4         System.out.println(result);
5     }
6 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\amitd\OneDrive\Desktop\Java> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\amitd\AppData\Roaming\Code\User\workspaceStorage\369c682bc15990d49b57e7e0f9168ddd\redhat.java\jdt_ws\Java_9d12eabf\bin' 'gc10'
2.9760461760461765
PS C:\Users\amitd\OneDrive\Desktop\Java>
```

Explain:

I learned how to use `double` for decimal calculations in Java. I learned how to evaluate complex mathematical expressions using arithmetic operators and parentheses.